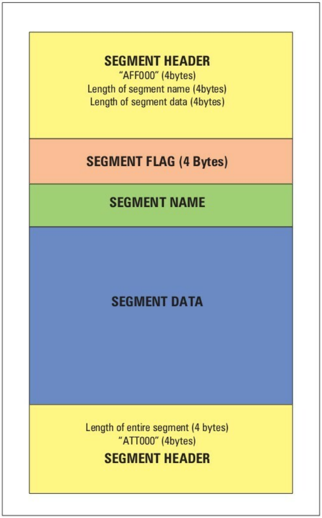

Figure 4: AFF file segment organisation

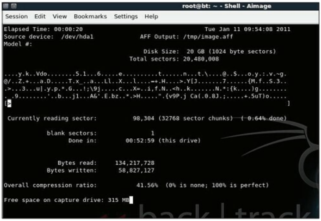

Figure 5: Real-time display of 'aimage'

resource requirements of extensive forensic analysis tools. Also, working with an image keeps the original media undisturbed and unaltered by any of the analysis activity. Here is a list of some of the important image-acquiring tools available in BackTrack.

Figure 6: Output of 'afinfo' command line utility

dd

'*dd*' is the popular Linux command-line utility for taking images of digital media, such as a hard drive, CD-ROM drive, USB drive, etc, or a particular partition of the media. Images created using '*dd*'-like utilities are called *raw* images, as these are bit-by-bit copies of the source media, without any additions or deletions. The basic usage for '*dd*' is:

```
dd if=<media/partition on a media> of=<image_file>
```

Examples include:

```
dd  if=/dev/hdc   of=image.dd
dd  if=/dev/hdc1  of=image1.dd
```

The first example takes the image of the whole disk (*hdc*), including the boot record and partition table. The second takes the image of a particular partition (*hdc1*) of the disk (*hdc*). A partition image can be mounted via the *loop* device. Also, instead of storing the image contents in a file, you can pipe the data to a host on the network by using tools such as '*nc* '; this is useful when you have space constraints for the image file. For example:

```
dd  if=/dev/hdc  |  nc  192.168.0.2  6000
```

In this example, the contents of the image file are sent via TCP port 6000 to the host *192.168.0.2*, where you can run '*nc*' in listening mode as '*nc -l 6000 > image .dd* ' to store the received image contents in the file *image.dd*. However, '*dd* ' has limitations in cases where the media to be imaged has errors in the form of bad sectors. Thus, a more advanced version of '*dd* ', named '*dd_rescue* ', is available in BackTrack to image a drive that's suspected to have one or more bad sectors.

dd_rescue

'*dd_rescue* ' is intended specifically for imaging potentially failing media. It switches to a smaller block size (essentially